Software Requirements Specification

Procurement Process Digitalization MVP — ProcureFlow

Field	Value
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Standard Basis	IEEE 830 / IEEE 29148

1. Introduction

1.1 Purpose

This SRS defines the functional and non-functional requirements for an MVP that digitalizes a manufacturing procurement workflow from **Purchase Request (PR)** creation through **Request for Quotation (RFQ)** drafting, vendor selection, RFQ issuance, and downloadable RFQ PDF output.

1.2 Scope

In scope:

- PR creation, editing (draft), submission, approval, and cancellation (from submitted state only).
- RFQ creation only from approved PR lines — one RFQ maximum per PR line.
- RFQ draft editing (dates, currency, terms, vendors), RFQ issue, RFQ cancel (draft only, admin).
- RFQ PDF download via DomPDF.
- Read-only master data views: products with BOM, raw materials, vendors with sourcing data.
- Role-based access control enforced at route, controller, and service layers.

Deferred — schema-ready, not exposed as end-user workflow:

- Recording vendor quotations in the UI (`quotation` table exists in schema).
- Award selection and audit fields.
- Automated outbound email with PDF attachment.

Out of scope:

- Purchase Order lifecycle.
- Inventory and accounting integration.
- Supplier portal and automated procurement optimization.

1.3 Definitions

Term	Definition
PR	Internal Purchase Request for required raw materials
RFQ	Request for Quotation sent to one or more vendors
BOM	Bill of Materials — raw material requirements per finished product
MVP	Minimum Viable Product
UoM	Unit of Measure
Snapshot	Fields copied verbatim from PR line at RFQ creation — immutable after issue

1.4 References

- Assignment 2: Procurement Process Digitalization MVP (AVERROA IT Launchpad).
- Repository: `README.md` , `Documentation/DATABASE.md` .
- IEEE 830 and IEEE 29148 guidance.

2. Overall Description

2.1 Business Domain

A manufacturing company purchases raw materials from international vendors. Master data — products, BOM, materials, vendors, currencies — supports planning and purchasing decisions. PR

and RFQ documents provide traceability and control over spending.

Core pain addressed: the manual procurement cycle caused delays, no audit trail, repetitive effort, and avoidable operational risk. This MVP establishes the first layer of digital control.

2.2 Core Workflow

1. A **requester** creates a PR in **draft** with one or more lines (material, quantity, unit, required delivery date).
2. The requester **submits** the PR → status becomes **submitted**.
3. A **procurement manager** reviews the PR and either **approves** it → **approved**, or **cancels** it from submitted → **cancelled** (cancellation reason required).
4. A **purchasing officer** creates an RFQ **draft** from an approved PR line, selects currency and vendors, and sets RFQ date and quotation due date.
5. The purchasing officer **issues** the RFQ → status becomes **issued** (issuer and timestamp recorded).
6. Any authenticated stakeholder **downloads** the RFQ PDF. Email automation is not implemented.

2.3 User Roles

Role	Code	Responsibilities
Requester	`requester`	Create, edit, and submit own PRs (draft only); view own PRs
Procurement Manager	`procurement_manager`	Approve or cancel submitted PRs; cannot approve own PR
Purchasing Officer	`purchasing_officer`	Create, edit, and issue RFQ drafts from approved PR lines
Admin	`admin`	All of the above; only role that can cancel a draft RFQ

Master data pages are not role-restricted — any authenticated user may view them.

2.4 Operating Environment

Component	Technology
Backend	Laravel 12, PHP 8.2+

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Database	MySQL
PDF Generation	barryvdh/laravel-dompdf
Frontend	Blade templates, Tailwind CSS, Alpine.js (Vite build)
Client	Modern web browser

3. System Features

3.1 Purchase Request Management

- PR header: auto-generated number (`PR-YYYYMMDD-XXXX`), requester, department snapshot, request date, status, notes.
- PR lines: material, quantity, unit, required delivery date, line notes.
- Lifecycle: `draft → submitted → approved` or `submitted → cancelled` .
- Audit fields: `approved_by` , `approved_at` , `cancelled_by` , `cancelled_at` , `cancellation_reason` .
- Approved PRs cannot be cancelled in the current service layer.

3.2 RFQ Management

- Strictly one RFQ per PR line — enforced by database unique constraint and form validation.
- Snapshot fields copied from PR line at RFQ creation: `material_id` , `quantity` , `unit_id` , `required_delivery_date` . These fields are immutable once the RFQ is issued.
- RFQ header fields: `rfq_date` , `quotation_due_date` , `currency_id` , payment terms, delivery location.
- Recipients: multiple vendors per RFQ; duplicate vendor per RFQ prevented by unique key.
- Issue transition: `draft → issued` ; records `issued_by` and `issued_at` .
- Cancel: **draft only**, **admin only**.

3.3 Vendor Quotation Management (Deferred)

- The `quotation` table supports `recipient_id` , `version_no` , `unit_price` , `lead_time_days` , `status` .
- No quotation create, edit, or view screens exist in the current application.

- Currency for quoted prices is not stored on `quotation`; commercial currency is defined on the parent RFQ via `currency_id`.

3.4 RFQ PDF Generation

- Any authenticated user downloads PDF via `GET /rfqs/{rfq}/pdf`.
- PDF content: RFQ number, PR reference, material snapshot, quantity, dates, recipients, commercial terms, vendor instructions.
- Generation is on-demand; `pdf_path` field is not required to persist.

3.5 Master Data (Read-Only Views)

- Products with BOM lines.
- Raw materials with base unit and vendor sourcing data (last price, currency, lead time, minimum order quantity).
- Vendors with vendor–material rows.
- No CRUD for master data in this MVP release.

3.6 Currency Management

- `currencies` table stores code (ISO 4217), name, and symbol.
- RFQ references one currency via `currency_id` — defines the required currency for all vendor responses.
- `vendor_materials.currency_id` records the currency of the last known price from that vendor.
- Quotation inherits currency from its parent RFQ; no separate currency column on `quotation`.

4. External Interface Requirements

4.1 User Interface

- PR list, create, edit, and show pages with actions gated by role and status.
- RFQ list, show, and edit pages; RFQ creation form embedded on the PR show page.
- RFQ PDF download action.
- Master data listing pages accessible to all authenticated users.

4.2 Software Interfaces

Interface	Purpose
Laravel application	HTTP routes, controllers, form requests, services
MySQL	Persistence, foreign keys, check constraints, unique constraints
DomPDF	RFQ PDF rendering from Blade template

4.3 Communication Interfaces

- HTTP/HTTPS only.
- Outbound email is not implemented; PDF is downloaded manually and attached by the user.

5. Functional Requirements

5.1 Implemented

ID	Requirement	Verification
FR-PR-001	The system shall assign a unique PR number at creation in the format `PR-YYYYMMDD-XXXX` (daily sequence, zero-padded to 4 digits).	Inspection / Test
FR-PR-002	The system shall require at least one PR line and a valid request date before accepting PR submission.	Test
FR-PR-003	The system shall reject PR lines with quantity ≤ 0.	Test
FR-PR-004	The system shall reject PR lines where required delivery date is before today.	Test

ID	Requirement	Verification
FR-PR-005	The system shall set PR initial status to **draft** at creation.	Inspection
FR-PR-006	PR status transitions: `draft → submitted`, `submitted → approved`, `submitted → cancelled`. Approved PRs cannot be cancelled.	Test
FR-PR-007	The system shall record `approved_by`, `approved_at`, `cancelled_by`, `cancelled_at`, and `cancellation_reason` on the PR header for audit purposes.	Inspection / Test
FR-PR-008	A procurement manager shall not be permitted to approve a PR they created.	Test
FR-RFQ-001	RFQ creation is allowed only for PR lines whose parent PR status is **approved**.	Test
FR-RFQ-002	At most one RFQ row per PR line (`rfq.pr_line_id` unique).	Test
FR-RFQ-003	RFQ shall snapshot `material_id`, `quantity`, `unit_id`, `required_delivery_date` from the PR line at creation.	Inspection
FR-RFQ-004	The system shall assign a unique RFQ number in the format `RFQ-YYYYMMDD-XXXX`.	Inspection
FR-RFQ-005	`quotation_due_date` shall be ≥ `rfq_date` (validated).	Test
FR-RFQ-006	RFQ shall require a `currency_id` referencing the `currencies` table.	Test

ID	Requirement	Verification
FR-RFQ-007	RFQ PDF shall include: RFQ number, PR reference, material, quantity, unit, RFQ date, quotation due date, currency, requester/department, vendor list, payment terms, delivery location, and vendor instructions.	Demonstration
FR-RFQ-008	RFQ issue transition (`draft → issued`) shall be performed only by `purchasing_officer` or `admin` .	Test
FR-RFQ-009	RFQ cancel shall be restricted to **draft** status and **admin** role only.	Test
FR-REC-001	Multiple vendors may be added as recipients per RFQ.	Test
FR-REC-002	A vendor may not be added twice to the same RFQ (unique `rfq_id` , `vendor_id`).	Test
FR-SEC-001	Role middleware and controller-level checks shall enforce all PR and RFQ permission rules at the server side, independent of UI rendering.	Test

6. Non-Functional Requirements

ID	Requirement	Notes
NFR-SEC-001	All business routes require authentication	Implemented via `auth` middleware
NFR-SEC-002	Role-based authorization on all sensitive actions	Implemented via `role` middleware and `abort(403)` in controllers/services
NFR-SEC-003	Passwords stored as hashes	Laravel default hashing (bcrypt)

ID	Requirement	Notes
NFR-DATA-001	Referential integrity enforced at database level	Foreign keys and check constraints on all domain tables
NFR-AUD-001	All transactional entities carry `created_at` and `updated_at`	Eloquent timestamps on all models
NFR-AUD-002	PR approval and cancellation are traceable to the acting user and timestamp	`approved_by`, `approved_at`, `cancelled_by`, `cancelled_at` on `pr_header`
NFR-PERF-001	Standard page load acceptable for LAN/intranet use	MVP target; not continuously measured
NFR-AVL-001	Availability dependent on deployment environment	Outside MVP scope

7. Data Model Overview

7.1 Core Entities

Master data:

`departments`, `users`, `currencies`, `units`, `products`, `raw_materials`, `bom`, `vendors`, `vendor_materials`

Transactional:

`pr_header`, `pr_lines`, `rfq`, `rfq_recipients`, `quotation` (UI deferred)

Framework:

`cache`, `sessions`, `jobs`, `password_reset_tokens` (Laravel defaults)

7.2 Key Relationships

- One **department** has many **users**.
- One **user** (requester) has many **PR headers**.
- One **PR header** has many **PR lines**.
- One **PR line** has zero or one **RFQ** (strict 1:1, enforced).

- One **RFQ** references one **currency** for all commercial terms.
- One **RFQ** has many **RFQ recipients** (one per vendor contacted).
- One **RFQ recipient** may have many **quotations** (schema only; UI deferred).
- One **vendor** has many **vendor materials**; each **vendor material** references a **currency** for `last_price`.

7.3 Key Design Decisions

- **Snapshot pattern:** RFQ copies demand fields from PR line at creation. Changes to PR line after RFQ creation do not affect the issued document.
- **Soft deletes:** applied to `pr_header` and `rfq` — business documents are never physically deleted.
- **Audit trail:** approval and cancellation actors and timestamps stored directly on `pr_header`.
- **Currency architecture:** defined once on `rfq`; `quotation` inherits it through the relationship chain, avoiding duplication.
- **Department snapshot:** PR header stores `department_id` at creation time — user department changes do not affect historical documents.

8. State Transitions

8.1 PR Lifecycle

Current State	Next State	Actor
draft	submitted	Requester / Admin
submitted	approved	Procurement Manager / Admin (cannot be own PR)
submitted	cancelled	Procurement Manager / Admin (reason required)
approved	*(no transition implemented)*	—
cancelled	terminal	—

8.2 RFQ Lifecycle

Current State	Next State	Actor
draft	issued	Purchasing Officer / Admin
draft	cancelled	Admin only
issued	*(reserved: closed, awarded)*	Not implemented
cancelled	terminal	—

Database enums include `closed` and `awarded` for forward compatibility. No application logic drives these transitions in the current release.

9. Assumptions and Constraints

- Master data is populated via seeders before any demo or testing session.
- All users have valid roles assigned in `users.role`.
- The MVP focuses on the PR → approval → RFQ draft → issue → PDF pipeline.
- Quotation capture and award are explicitly deferred though partially modeled in schema.
- Email automation is deferred to a future phase.

10. Traceability

Business Objective	Supporting Requirements
BO-01 — Digitize PR creation and approval	FR-PR-001 to FR-PR-008, FR-SEC-001
BO-02 — RFQ drafting, issuance, and PDF output	FR-RFQ-001 to FR-RFQ-009, FR-REC-001, FR-REC-002
BO-03 — Vendor and currency management	FR-RFQ-006, master data views
BO-04 — Quotation and award	Deferred (FR-QTN-001, FR-AWD-001)

Verification legend:

- **Test** — executable behaviour verified via browser or automated test.
- **Inspection** — code or schema review.
- **Demonstration** — stakeholder-visible output (e.g., PDF).

11. Appendix

11.1 Minimum RFQ PDF Content

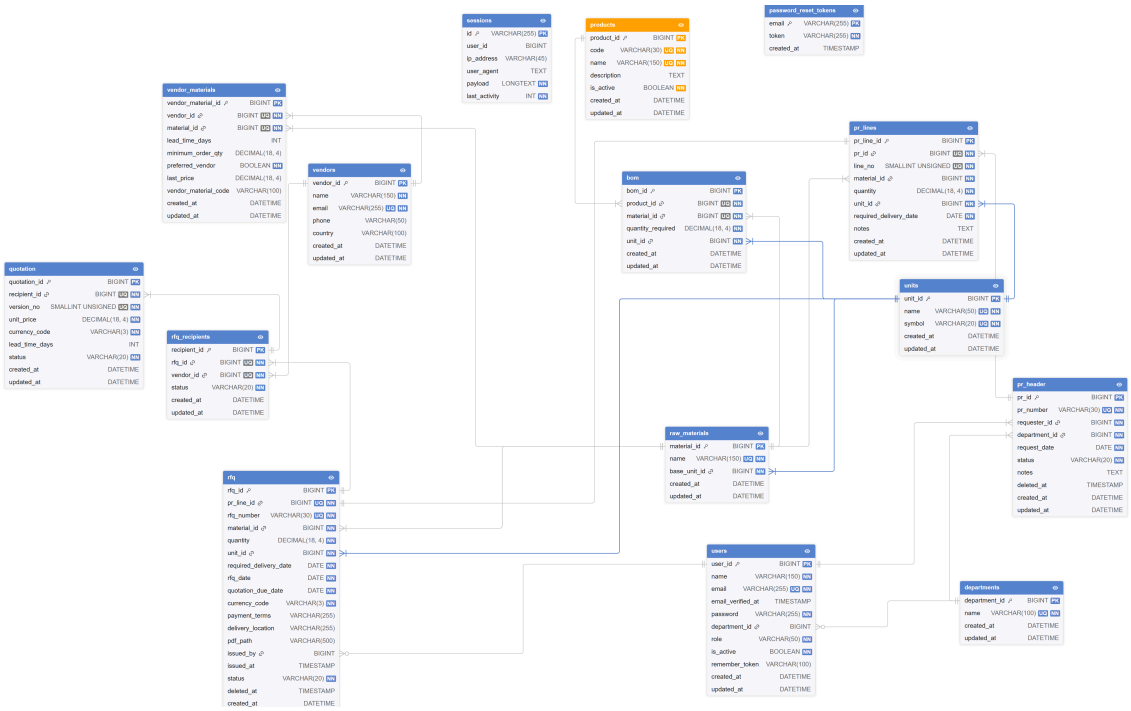
RFQ number · PR reference · material name · quantity and unit · RFQ date · quotation due date · currency (code and symbol) · requester name and department · vendor list (name, country, email) · payment terms · delivery location · instructions for vendor.

11.2 Default Demo Accounts

Role	Email	Password
Requester	ahmad@factory.com	password
Procurement Manager	sara@factory.com	password
Purchasing Officer	khalid@factory.com	password
Admin	admin@factory.com	password

11.3 Entity-Relationship Diagram (ERD)

The figure below is the **Entity-Relationship Diagram (ERD)** for the procurement database as implemented in this MVP (master data, PR/RFQ documents, and related links). This diagram shows entities and their relationships at a glance.


sessions
id VARCHAR(255) PK
user_id BIGINT
ip_address VARCHAR(45)
user_agent TEXT
payload LONGTEXT NN
last_activity INT NN
products
product_id BIGINT PK
code VARCHAR(30) UQ NN
name VARCHAR(150) UQ NN
description TEXT
is_active BOOLEAN NN
created_at DATETIME
updated_at DATETIME
password_reset_tokens
email VARCHAR(255) PK
token VARCHAR(255) NN
created_at TIMESTAMP
vendor_materials
vendor_material_id BIGINT PK
vendor_id BIGINT UQ NN
material_id BIGINT UQ NN
lead_time_days INT
minimum_order_qty DECIMAL(18, 4)
preferred_vendor BOOLEAN NN
last_price DECIMAL(18, 4)
vendor_material_code VARCHAR(100)
created_at DATETIME
updated_at DATETIME
vendors
vendor_id BIGINT PK
name VARCHAR(150) NN
email VARCHAR(255) UQ NN
phone VARCHAR(50)
country VARCHAR(100)
created_at DATETIME
updated_at DATETIME
pr_lines
pr_line_id BIGINT PK
pr_id BIGINT UQ NN
line_no SMALLINT UNSIGNED UQ NN
material_id BIGINT NN
quantity DECIMAL(18, 4) NN
unit_id BIGINT NN
required_delivery_date DATE NN
notes TEXT
created_at DATETIME
updated_at DATETIME
bom
bom_id BIGINT PK
product_id BIGINT UQ NN
material_id BIGINT UQ NN
quantity_required DECIMAL(18, 4) NN
unit_id BIGINT NN
created_at DATETIME
updated_at DATETIME
quotation
quotation_id BIGINT PK
recipient_id BIGINT UQ NN
version_no SMALLINT UNSIGNED UQ NN
unit_price DECIMAL(18, 4) NN
currency_code VARCHAR(3) NN
lead_time_days INT
status VARCHAR(20) NN
created_at DATETIME
updated_at DATETIME
units
unit_id BIGINT PK
name VARCHAR(50) UQ NN
symbol VARCHAR(20) UQ NN
created_at DATETIME
updated_at DATETIME
rfq_recipients
recipient_id BIGINT PK
rfq_id BIGINT UQ NN
vendor_id BIGINT UQ NN
status VARCHAR(20) NN
created_at DATETIME
updated_at DATETIME
pr_header
pr_id BIGINT PK
pr_number VARCHAR(30) UQ NN
requester_id BIGINT NN
department_id BIGINT NN
request_date DATE NN
status VARCHAR(20) NN
notes TEXT
deleted_at TIMESTAMP
created_at DATETIME
updated_at DATETIME
raw_materials
material_id BIGINT PK
name VARCHAR(150) UQ NN
base_unit_id BIGINT NN
created_at DATETIME
updated_at DATETIME
rfq
rfq_id BIGINT PK
pr_line_id BIGINT UQ NN
rfq_number VARCHAR(30) UQ NN
material_id BIGINT NN
quantity DECIMAL(18, 4) NN
unit_id BIGINT NN
required_delivery_date DATE NN
rfq_date DATE NN
quotation_due_date DATE NN
currency_code VARCHAR(3) NN
payment_terms VARCHAR(255)
delivery_location VARCHAR(255)
pdf_path VARCHAR(500)
issued_by BIGINT
issued_at TIMESTAMP
status VARCHAR(20) NN
deleted_at TIMESTAMP
created_at DATETIME
users
user_id BIGINT PK
name VARCHAR(150) NN
email VARCHAR(255) UQ NN
email_verified_at TIMESTAMP
password VARCHAR(255) NN
department_id BIGINT
role VARCHAR(50) NN
is_active BOOLEAN NN
remember_token VARCHAR(100)
created_at DATETIME
updated_at DATETIME
departments
department_id BIGINT PK
name VARCHAR(100) UQ NN
created_at DATETIME
updated_at DATETIME